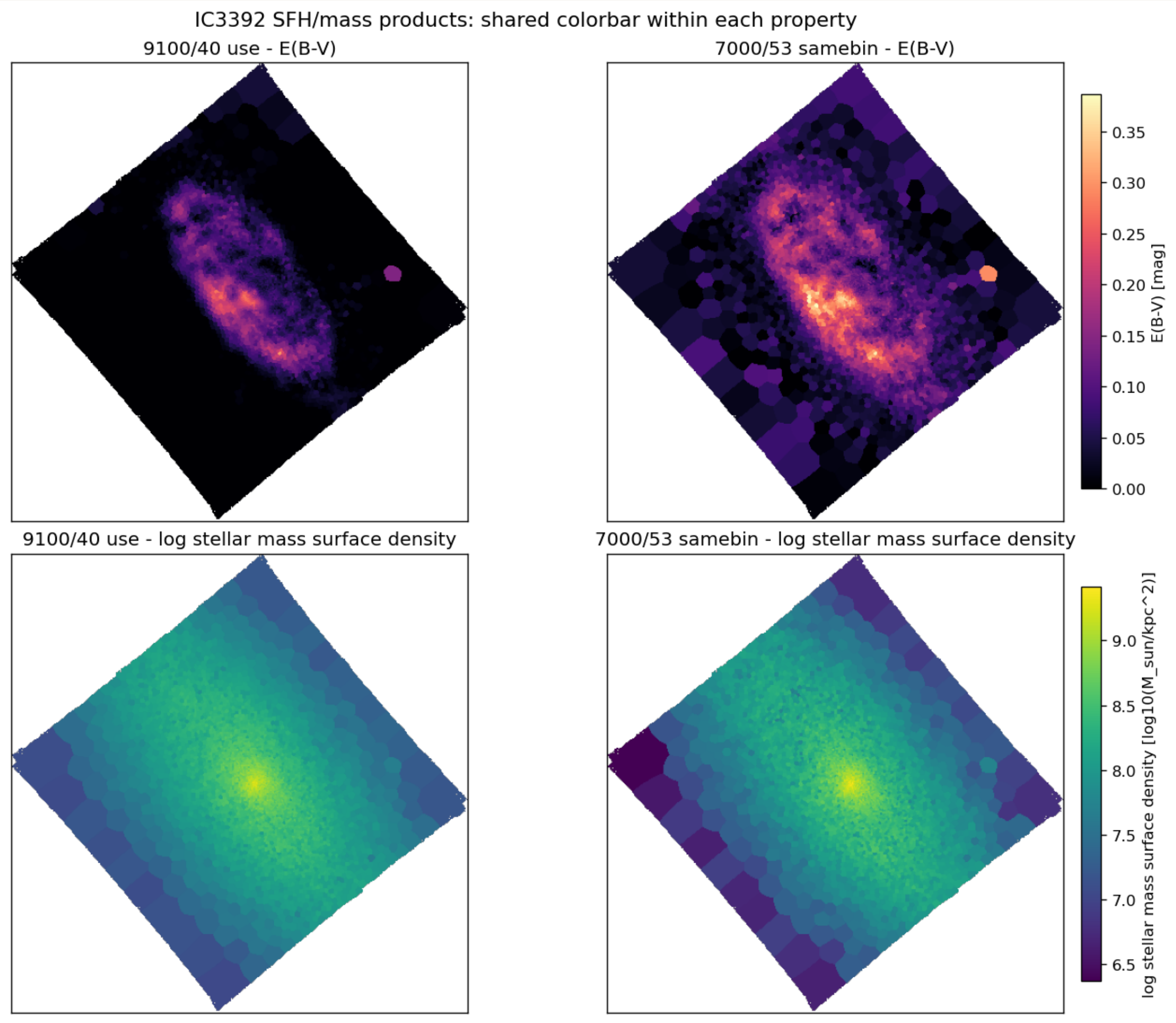
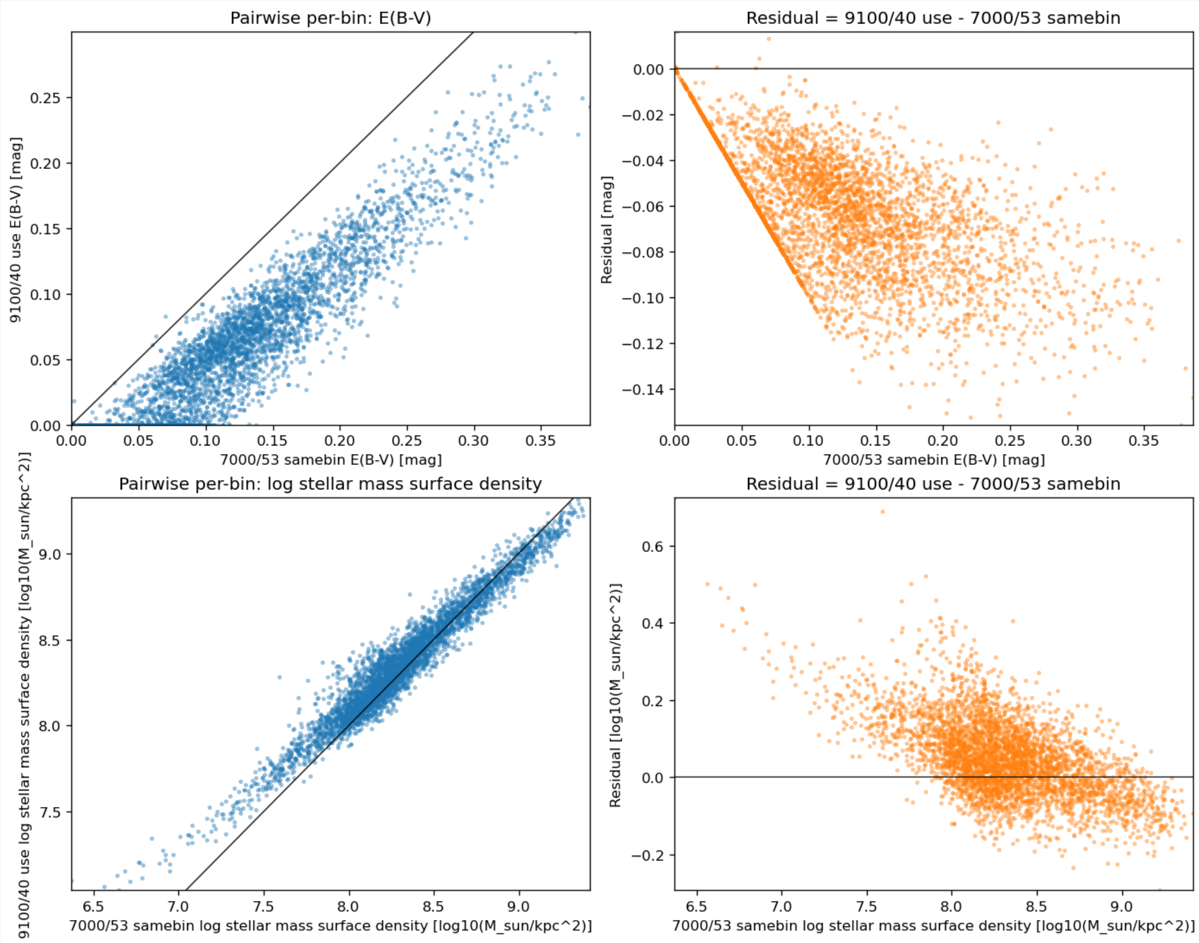
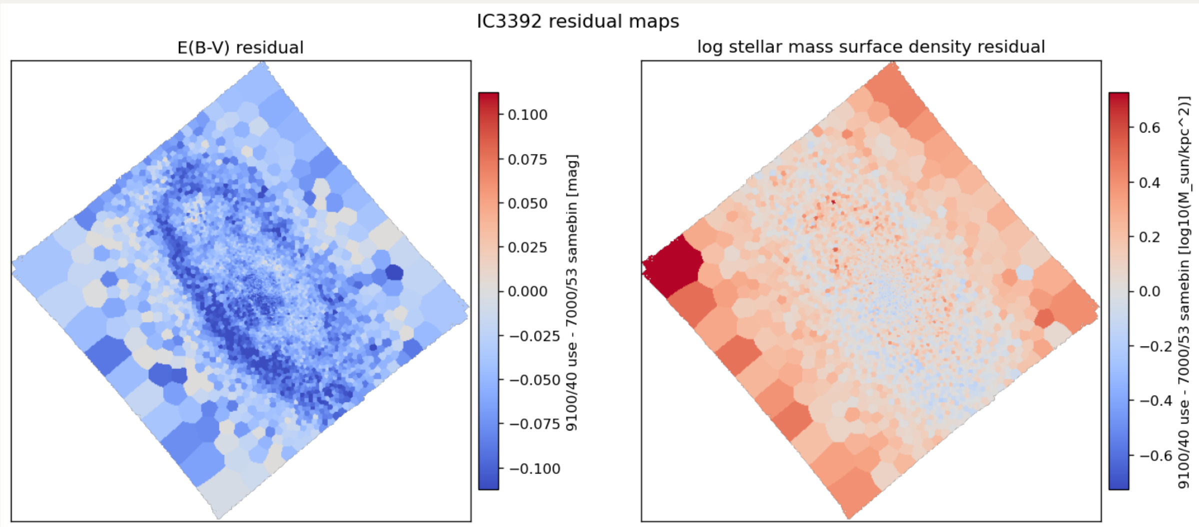
20260518 SFH and Stellar Mass Check

1. IC3392

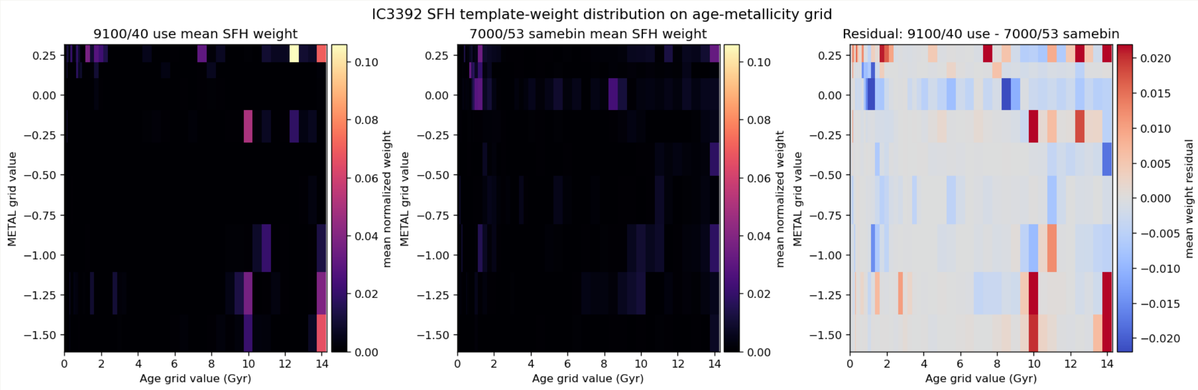
1.1 IC3392 E(B-V) and stellar mass maps.

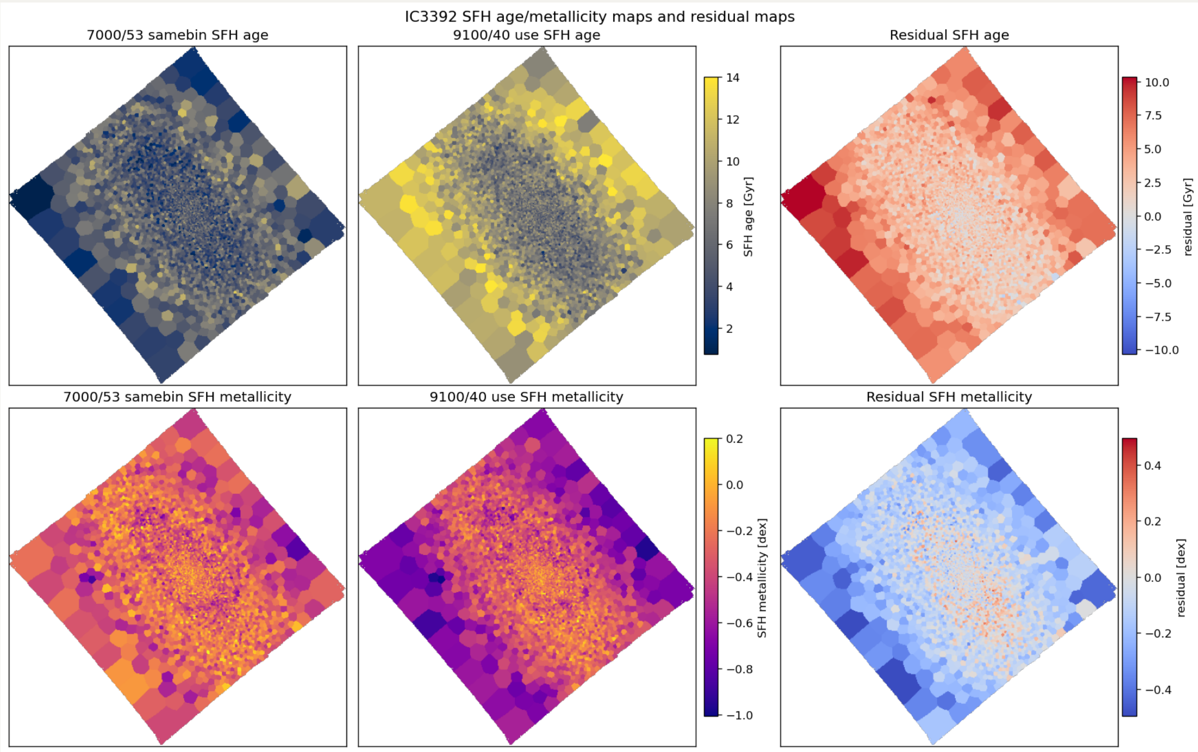


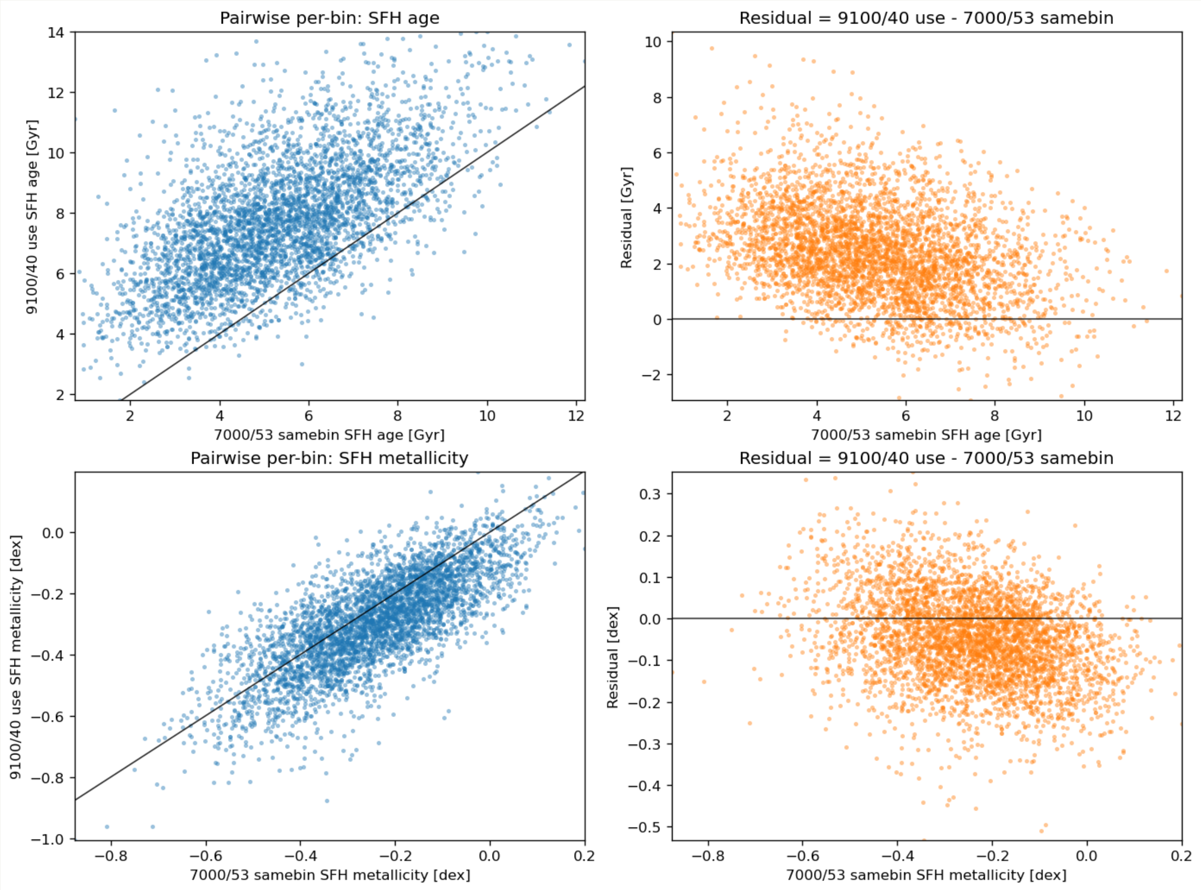




1.2 IC3392 SFH weights: Age and stellar metallicity [M/H]


IC3392 SFH template-weight distribution on age-metallicity grid
9100/40 use mean SFH weight
7000/53 samebin mean SFH weight
Residual: 9100/40 use - 7000/53 samebin
METAL grid value
Age grid value (Gyr)
mean normalized weight
mean weight residual


IC3392 SFH age/metallicity maps and residual maps
7000/53 samebin SFH age
9100/40 use SFH age
Residual SFH age
SFH age [Gyr]
residual [Gyr]
7000/53 samebin SFH metallicity
9100/40 use SFH metallicity
Residual SFH metallicity
SFH metallicity [dex]
residual [dex]



1.3 IC3392 Integrated values comparison

	QUANTITY	7000/53 SAMEBIN	9100/40 USE	9100/40 USE - 7000/53 SAMEBIN	UNIT
0	Total R-band magnitude (uncorrected)	12.2600	12.2600	0.000	mag AB
1	Total R-band magnitude (corrected)	11.8650	12.0770	0.212	mag AB
2	Total R-band luminosity	9.5330	9.4480	-0.085	log10(Lsun)
3	Total stellar mass (R)	9.7210	9.7790	0.058	log10(Msun)
4	Integrated stellar M/L_R	1.5402	2.1402	0.600	Msun/Lsun

1.4 IC3392 Sumary

For `E(B-V)`, the two maps have very similar spatial structure, but the `9100/40 use` run is systematically lower than the `7000/53 samebin` run: the median value shifts from `0.1168 mag` in `7000/53 samebin` to `0.0560 mag` in `9100/40 use`. Almost all common bins move downward: `3986 / 4032` bins have negative residuals, only `7` bins are positive, and `39` are exactly zero.

For `LOGMASS_SURFACE_DENSITY`, the median surface density changes from `8.2787 dex` in `7000/53 samebin` to `8.3433 dex` in `9100/40 use`.

The uncorrected total R-band magnitude is identical in the two logs, `12.260 AB`. After extinction correction, however, `9100/40 use` is fainter: `12.077 AB` compared with `11.865 AB`, giving a lower total R-band luminosity by `0.085 dex`.

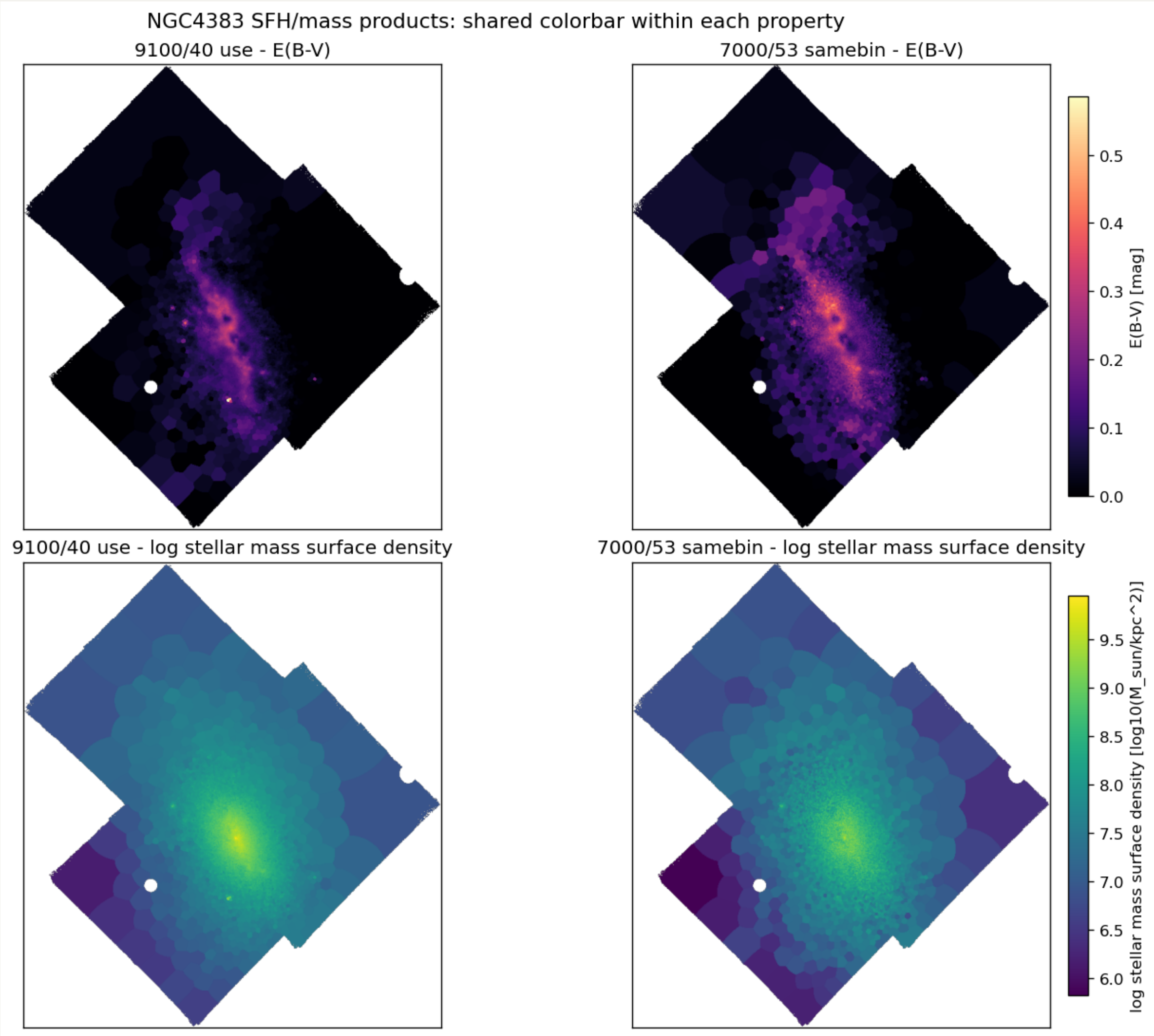
The total stellar mass is higher in `9100/40 use` by `0.058 dex`, because the integrated `M/L_R` is much higher: `2.1402` compared with `1.5402 M_sun/L_sun`.

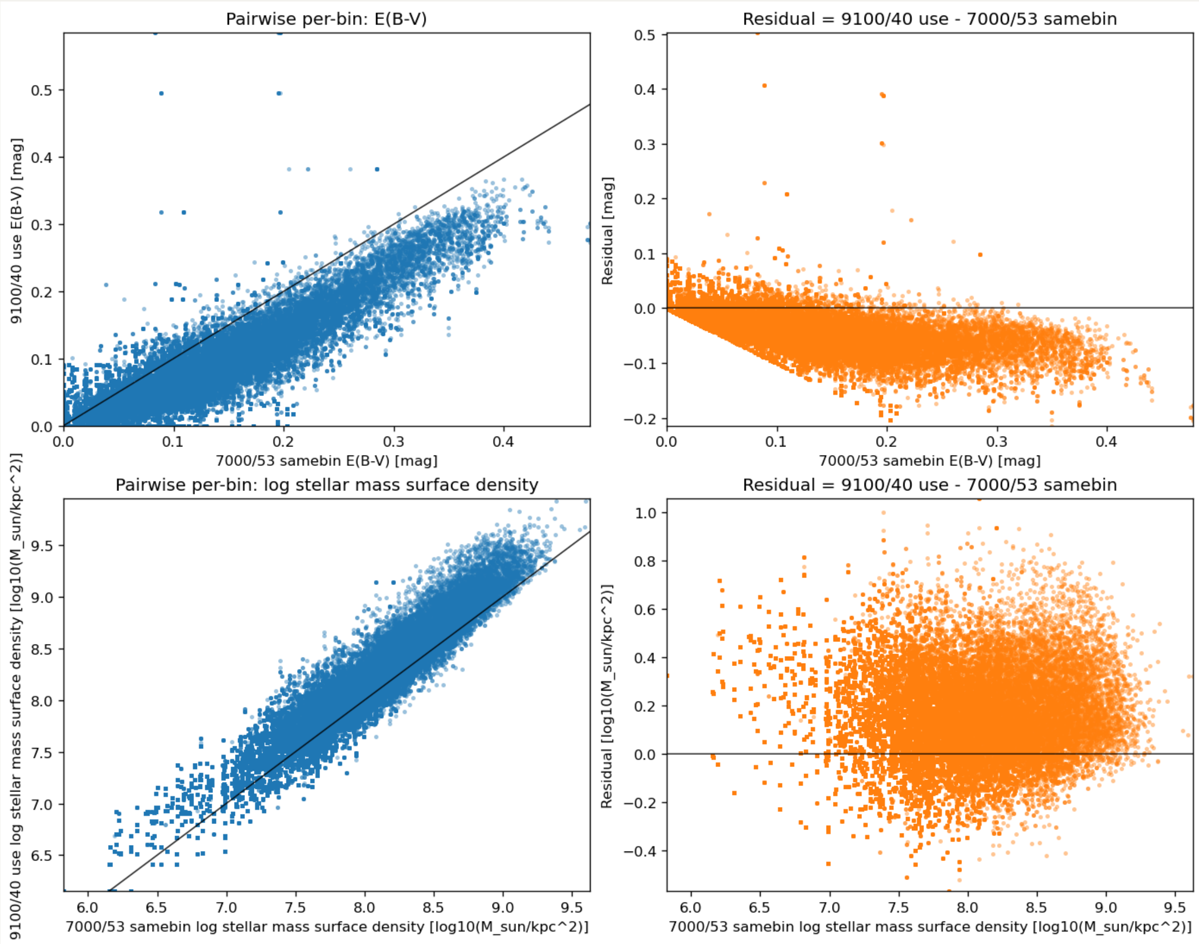
In SFH-weight-grid check, the `9100/40 use` run shifts toward older templates: the mean linear age from the weight grid increases from `5.321 Gyr` to `7.750 Gyr`, and the fraction of weight at ages `>= 8 Gyr` increases from `0.367` to `0.570`.

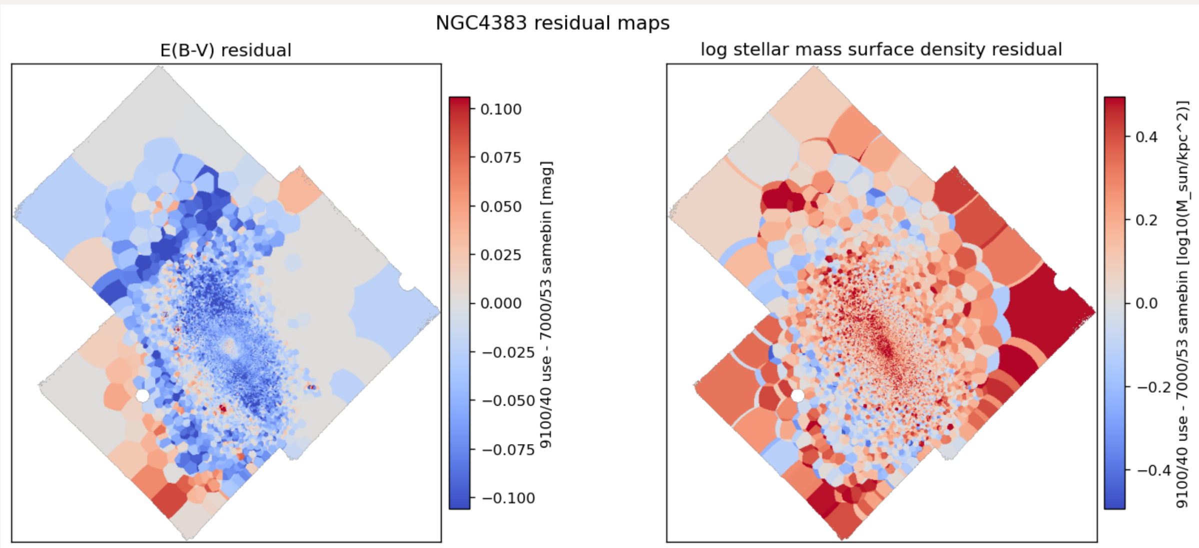
The spatial `AGE` and `METAL` maps show the same trend. The median fitted age increases from `5.242 Gyr` in `7000/53 samebin` to `7.638 Gyr` in `9100/40 use`, with a median residual of `+2.324 Gyr`; `3770 / 4032` bins are older in the `9100/40 use` run. The median metallicity decreases from `-0.238 dex` to `-0.286 dex`, with a median residual of `-0.0498 dex`; `2757 / 4032` bins are more metal-poor in the `9100/40 use` run. Age is therefore the stronger coherent shift, while metallicity also moves slightly lower.

2. NGC4383

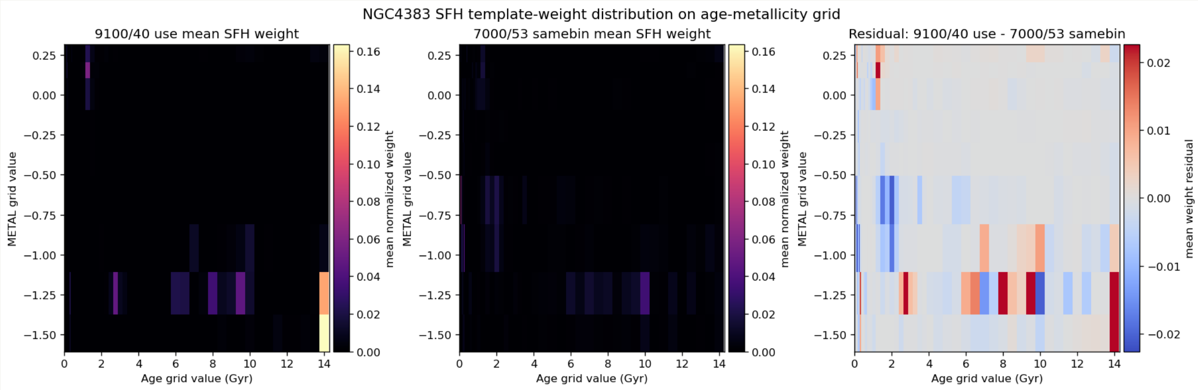
2.1 NGC4383 E(B-V) and stellar mass maps.

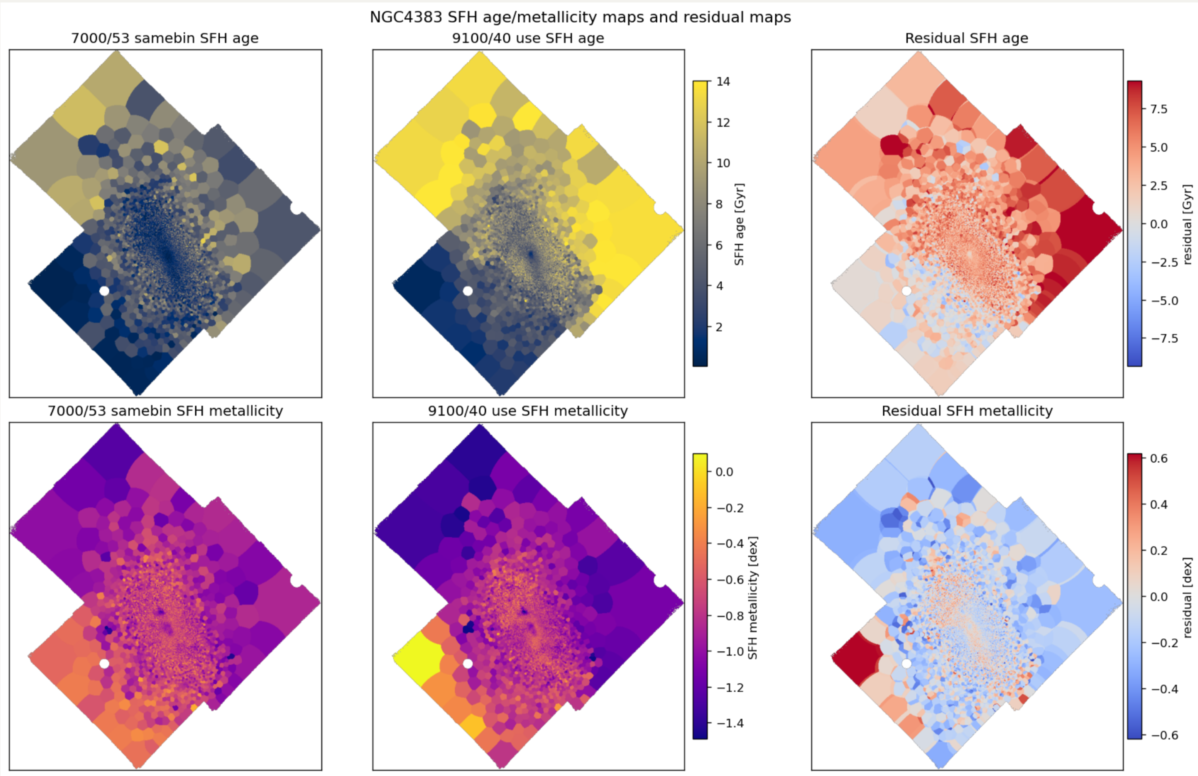

NGC4383 SFH/mass products: shared colorbar within each property
9100/40 use - E(B-V)
7000/53 samebin - E(B-V)
E(B-V) [mag]
0.0
0.1
0.2
0.3
0.4
0.5
9100/40 use - log stellar mass surface density
7000/53 samebin - log stellar mass surface density
log stellar mass surface density [log10(M_sun/kpc^2)]
6.0
6.5
7.0
7.5
8.0
8.5
9.0
9.5

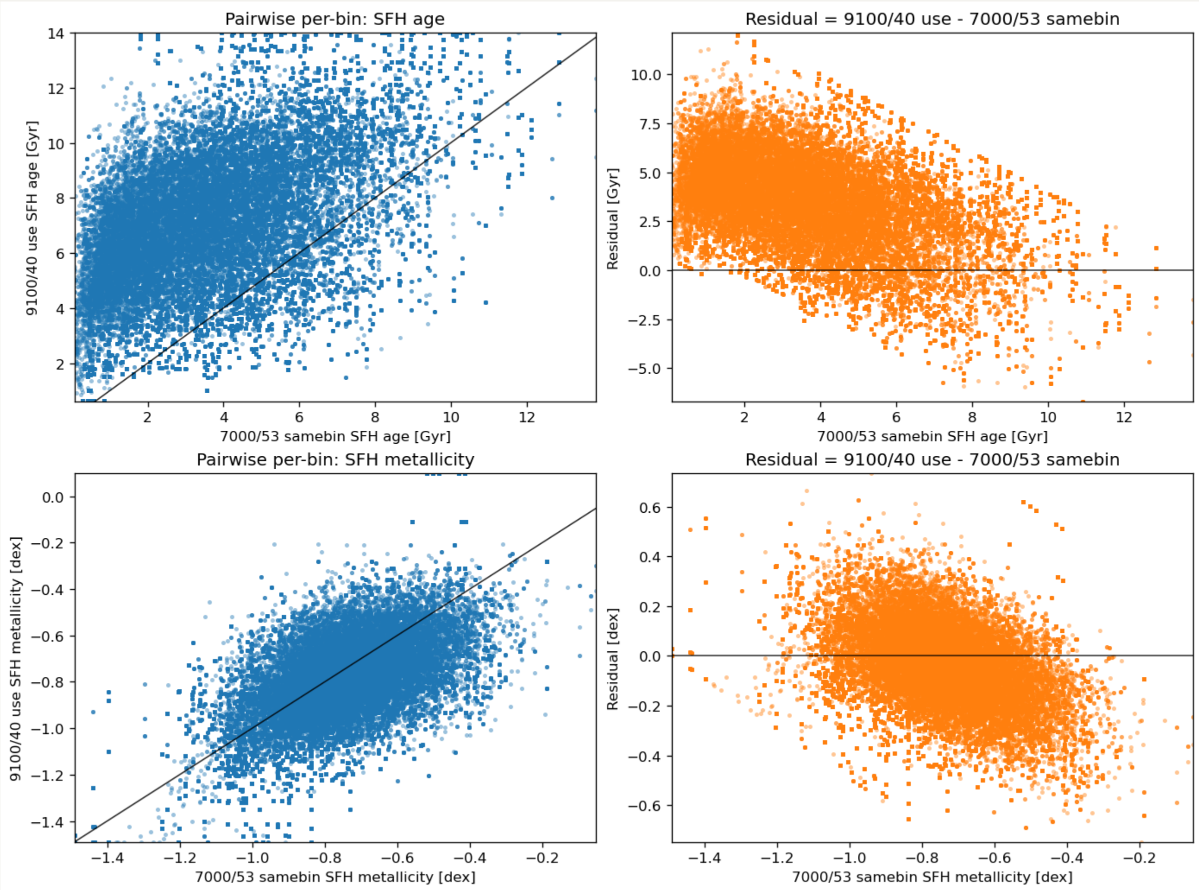




2.2 NGC4383 SFH weights: Age and stellar metallicity [M/H]


NGC4383 SFH template-weight distribution on age-metallicity grid
9100/40 use mean SFH weight
7000/53 samebin mean SFH weight
Residual: 9100/40 use - 7000/53 samebin
METAL grid value
Age grid value (Gyr)
mean normalized weight
mean weight residual


NGC4383 SFH age/metallicity maps and residual maps
7000/53 samebin SFH age
9100/40 use SFH age
Residual SFH age
SFH age [Gyr]
residual [Gyr]
7000/53 samebin SFH metallicity
9100/40 use SFH metallicity
Residual SFH metallicity
SFH metallicity [dex]
residual [dex]



2.3 NGC4383 Integrated values comparison

For `E(B-V)`, the spatial pattern is broadly preserved but the `9100/40 use` run is lower overall. The median value changes from `0.0249 mag` in `7000/53 samebin` to `0.0243 mag` in `9100/40 use`.

For `LOGMASS_SURFACE_DENSITY`, the median changes from `7.0749 dex` to `7.2259 dex`.

The parsed mass logs show that the uncorrected total R-band magnitude is unchanged at `11.842 AB`. After extinction correction, `9100/40 use` is fainter by `0.143 mag`, corresponding to a total R-band luminosity change of `-0.057 dex`.

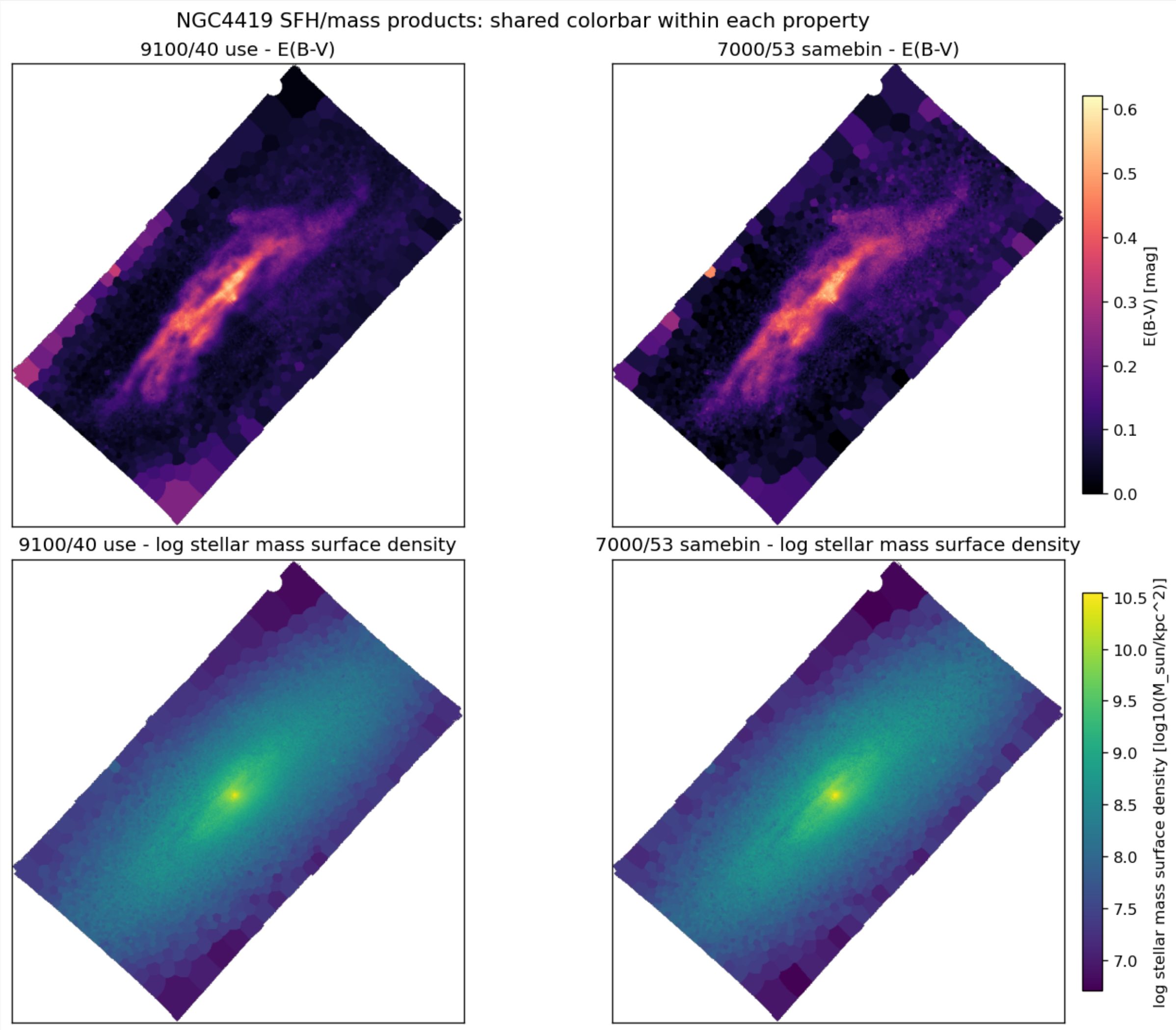
The total stellar mass changes by `0.178 dex`, and the integrated `M/L_R` changes from `0.7279` to `1.2494 M_sun/L_sun`.

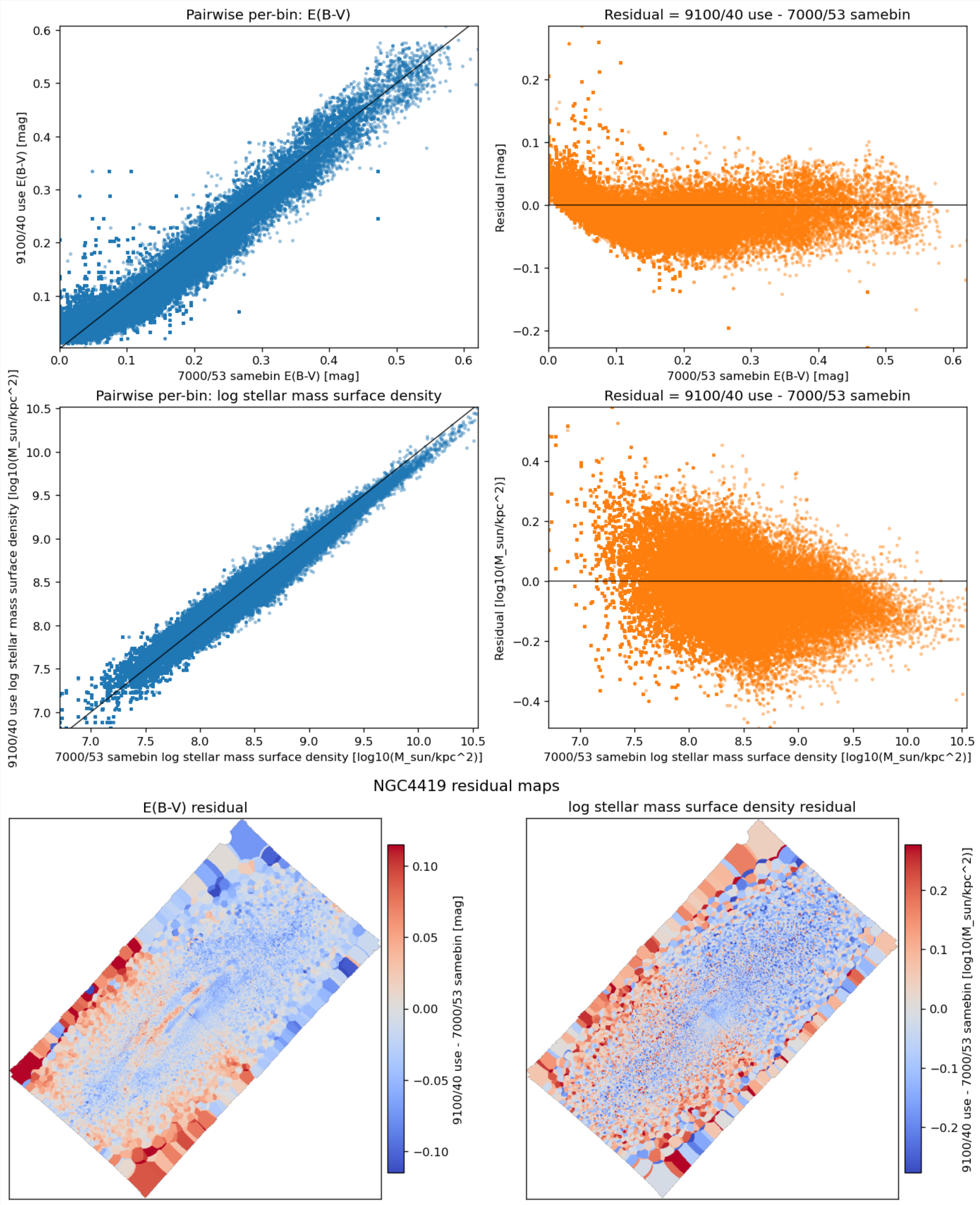
In SFH weight distribution, the mean age changes from `3.022 Gyr` to `6.822 Gyr`, and the weight fraction at ages `>= 8 Gyr` changes from `0.168` to `0.450`.

The spatial `AGE` and `METAL` products provide the map-level view of the same SFH changes. The median age changes from `4.986 Gyr` to `10.354 Gyr`. The median metallicity changes from `-0.8717 dex` to `-0.9836 dex`.

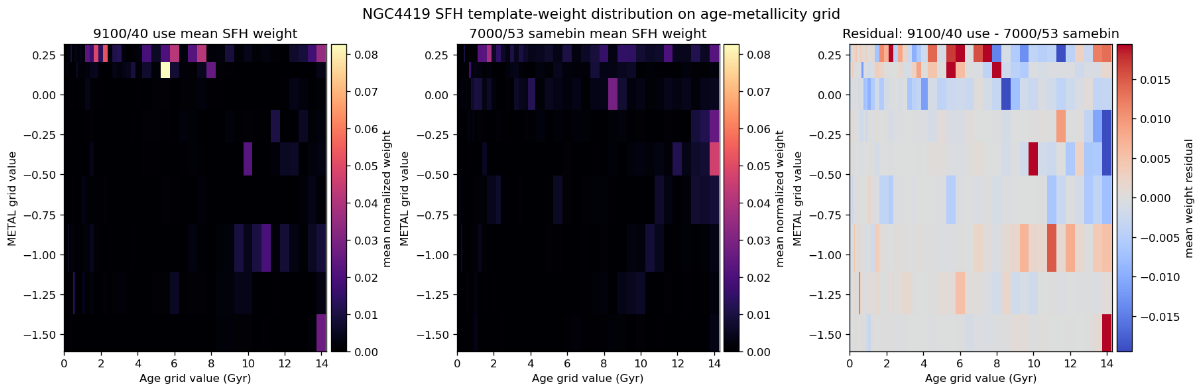
3. NGC4419

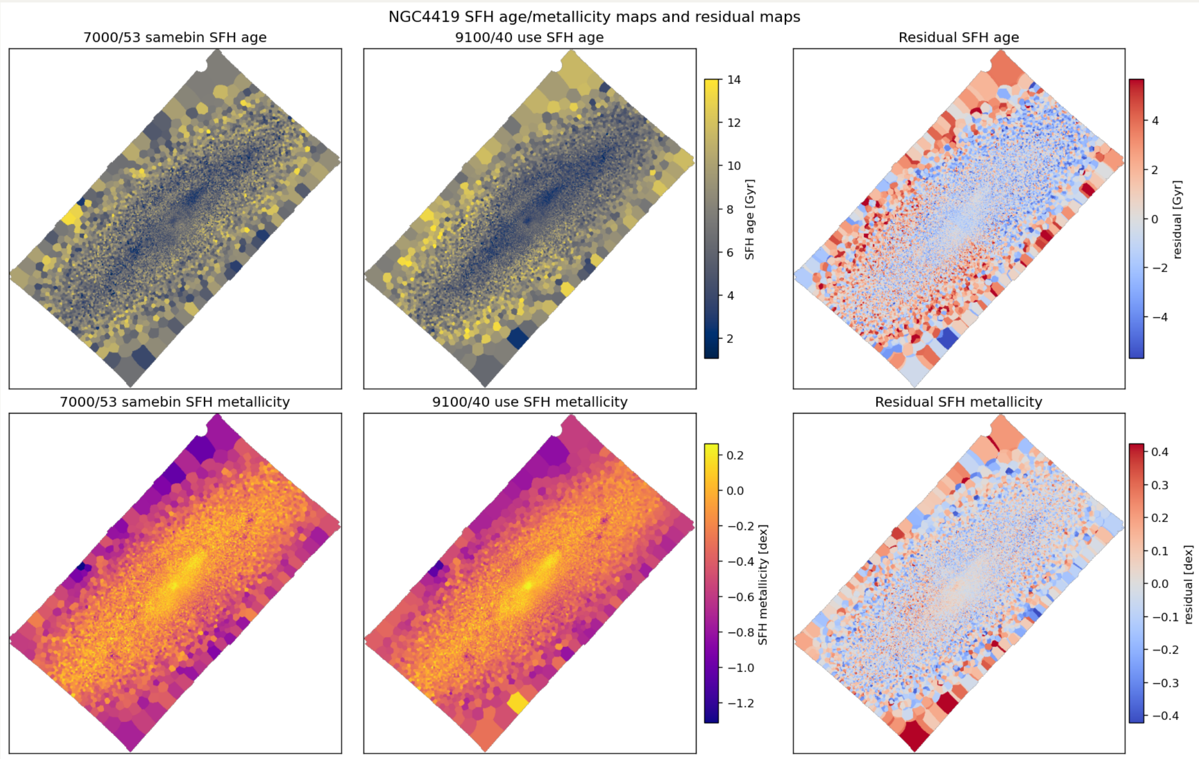
3.1 NGC4419 E(B-V) and stellar mass maps.

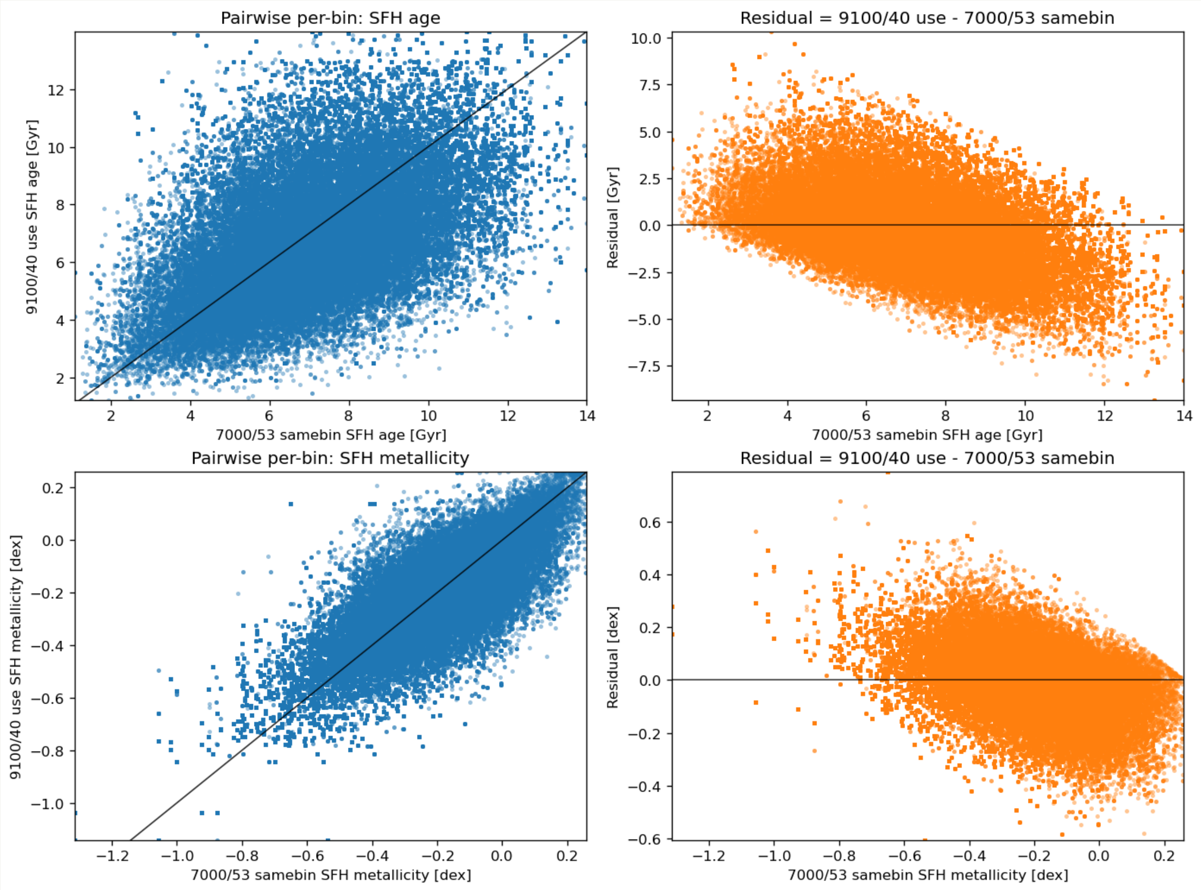




3.2 NGC4419 SFH weights: Age and stellar metallicity [M/H]


NGC4419 SFH template-weight distribution on age-metallicity grid
9100/40 use mean SFH weight
7000/53 samebin mean SFH weight
Residual: 9100/40 use - 7000/53 samebin
METAL grid value
Age grid value (Gyr)
mean normalized weight
mean weight residual


NGC4419 SFH age/metallicity maps and residual maps
7000/53 samebin SFH age
9100/40 use SFH age
Residual SFH age
SFH age [Gyr]
residual [Gyr]
7000/53 samebin SFH metallicity
9100/40 use SFH metallicity
Residual SFH metallicity
SFH metallicity [dex]
residual [dex]



3.3 NGC4419 Integrated values comparison

	QUANTITY	7000/53 SAMEBIN	9100/40 USE	9100/40 USE - 7000/53 SAMEBIN	UNIT
0	Total R-band magnitude (uncorrected)	10.8590	10.8590	0.0000	mag AB
1	Total R-band magnitude (corrected)	10.1380	10.2030	0.0650	mag AB
2	Total R-band luminosity	10.2240	10.1980	-0.0260	log10(Lsun)
3	Total stellar mass (R)	10.5090	10.4540	-0.0550	log10(Msun)
4	Integrated stellar M/L_R	1.9295	1.8032	-0.1263	Msun/Lsun

3.4 NGC4419 Sumary

For `E(B-V)`, the spatial pattern is broadly preserved but the `9100/40 use` run is lower overall. The median value changes from `0.0896 mag` in `7000/53 samebin` to `0.0754 mag` in `9100/40 use`.

For `LOGMASS_SURFACE_DENSITY`, the median changes from `7.8453 dex` to `7.8497 dex`.

The parsed mass logs show that the uncorrected total R-band magnitude is unchanged at `10.859 AB`. After extinction correction, `9100/40 use` is fainter by `0.065 mag`, corresponding to a total R-band luminosity change of `-0.026 dex`. The total stellar mass changes by `-0.055 dex`, and the integrated `M/L_R` changes from `1.9295` to `1.8032 M_sun/L_sun`.

In SFH weight grids: the mean age changes from `6.608 Gyr` to `6.064 Gyr`, and the weight fraction at ages `>= 8 Gyr` changes from `0.431` to `0.293`.

The spatial `AGE` and `METAL` products provide the map-level view of the same SFH changes. The median age changes from `7.759 Gyr` to `8.061 Gyr`. The median metallicity changes from `-0.3495 dex` to `-0.3435 dex`.